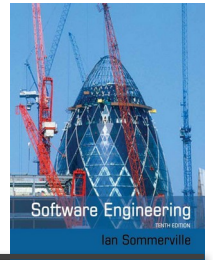


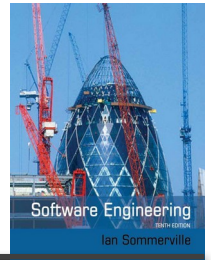
Chapter 2 – Software Processes

Topics covered



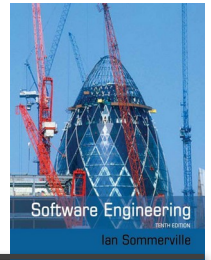
- ✧ Software process models
- ✧ Process activities

The software process



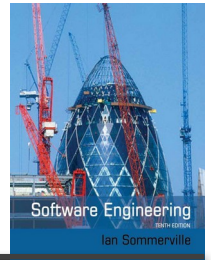
- ✧ A structured **set of activities** required to develop a software system.
- ✧ Many different software processes but all involve:
 - **Specification** – defining what the system should do;
 - **Design and implementation** – defining the organization of the system and implementing the system;
 - **Validation** – checking that it does what the customer wants;
 - **Evolution** – changing the system in response to changing customer needs.
- ✧ A software process model is an abstract representation of a process. It presents a description of a process from some particular perspective.

Software process descriptions



- ✧ When we describe and discuss processes, we usually talk about the activities in these processes such as specifying a data model, designing a user interface, etc. and the ordering of these activities.
- ✧ Process descriptions may also include:
 - Products, which are the outcomes of a process activity;
 - Roles, which reflect the responsibilities of the people involved in the process;
 - Pre- and post-conditions, which are statements that are true before and after a process activity has been enacted or a product produced.

Plan-driven and agile processes

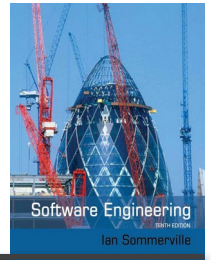


- ✧ **Plan-driven processes** are processes where all of the process activities are planned in advance and progress is measured against this plan.
- ✧ **In agile processes**, planning is incremental and it is easier to change the process to reflect changing customer requirements.
- ✧ In practice, most practical processes include elements of both plan-driven and agile approaches.
- ✧ **There are no right or wrong software processes.**



Software process models

Software process models



✧ The waterfall model

- Plan-driven model. Separate and distinct phases of specification and development.

✧ Incremental development

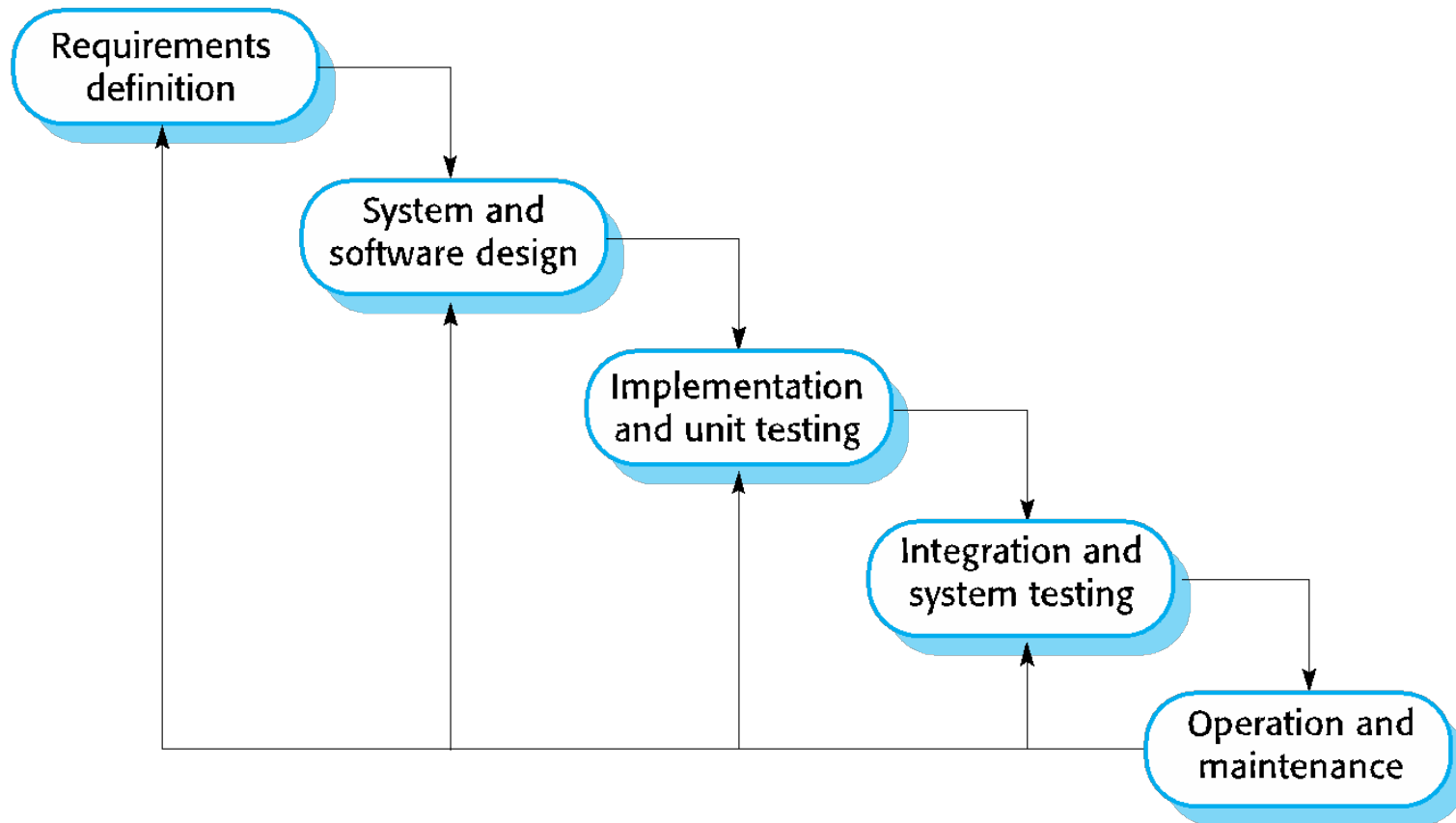
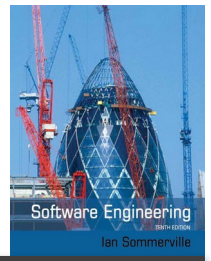
- Specification, development and validation are interleaved. May be plan-driven or agile.

✧ Reuse-oriented software engineering

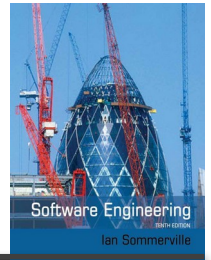
- The system is assembled from existing components. May be plan-driven or agile.

✧ In practice, most large systems are developed using a process that incorporates elements from all of these models.

The waterfall model

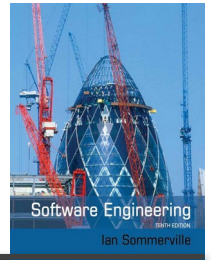


Waterfall model phases



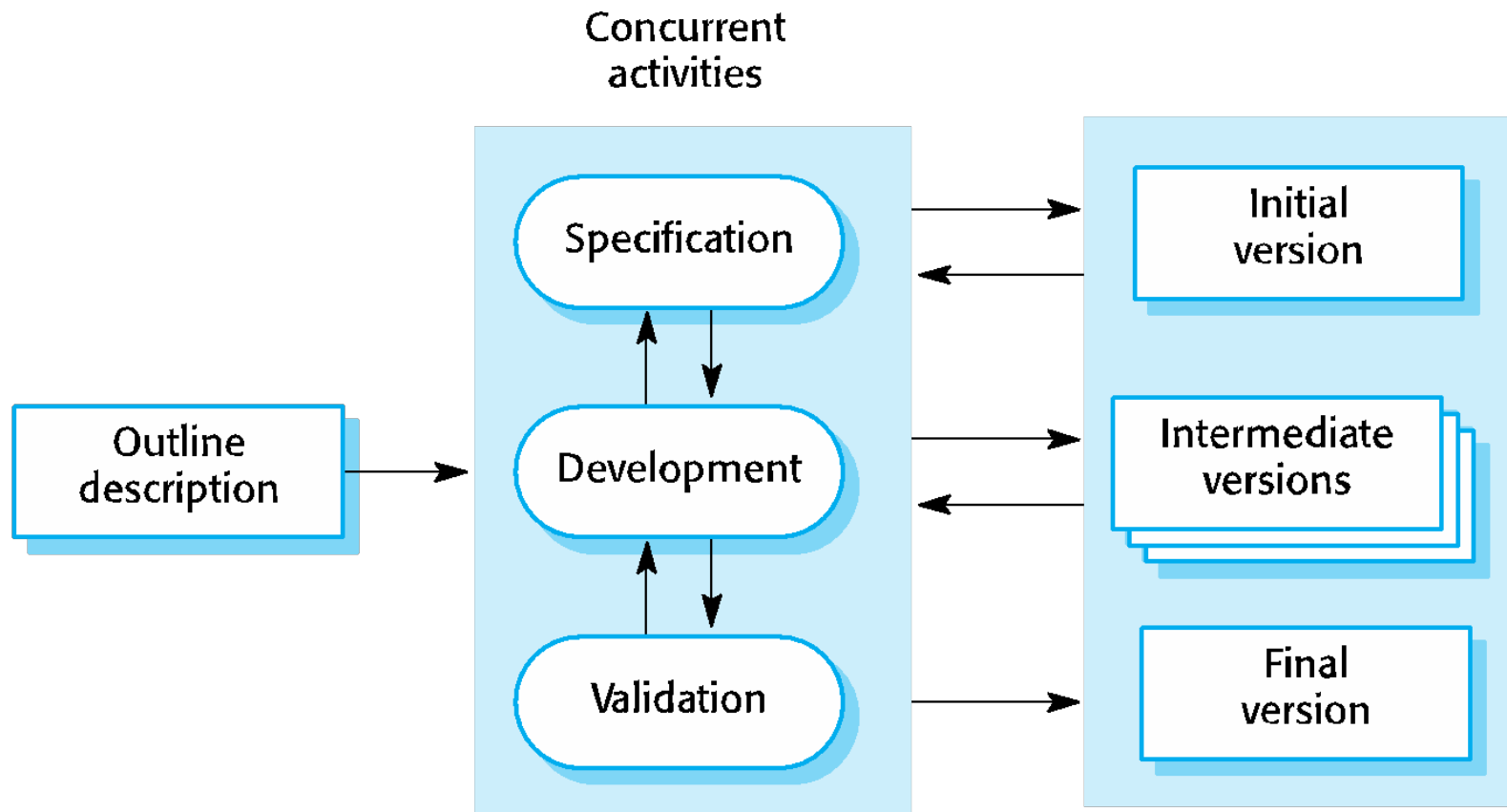
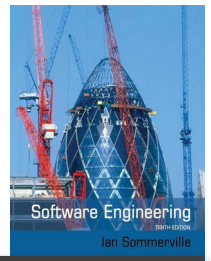
- ✧ There are separate identified phases in the waterfall model:
 - Requirements analysis and definition
 - System and software design
 - Implementation and unit testing
 - Integration and system testing
 - Operation and maintenance
- ✧ The main drawback of the waterfall model is the difficulty of accommodating change after the process is underway. In principle, a phase has to be complete before moving onto the next phase.

Waterfall model problems

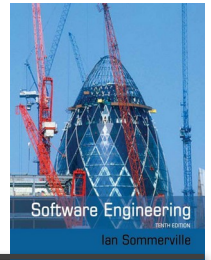


- ✧ **Inflexible** partitioning of the project into distinct stages makes it **difficult to respond to changing customer requirements**.
 - Therefore, this model is only appropriate when the requirements are well-understood and changes will be fairly limited during the design process.
 - Few business systems have stable requirements.
- ✧ The waterfall model is mostly used for **large systems engineering projects** where a system is developed at several sites.
 - In those circumstances, the plan-driven nature of the waterfall model helps coordinate the work.

Incremental development



Incremental development benefits

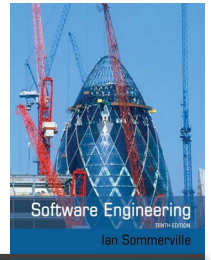


- ✧ **The cost of accommodating changing customer requirements is reduced.**
 - The amount of analysis and documentation that has to be redone is much less than is required with the waterfall model.
- ✧ **It is easier to get customer feedback on the development work that has been done.**
 - Customers can comment on demonstrations of the software and see how much has been implemented.
- ✧ **More rapid delivery and deployment of useful software to the customer is possible.**
 - Customers are able to use and gain value from the software earlier than is possible with a waterfall process.



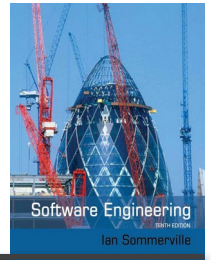
Process activities

Process activities



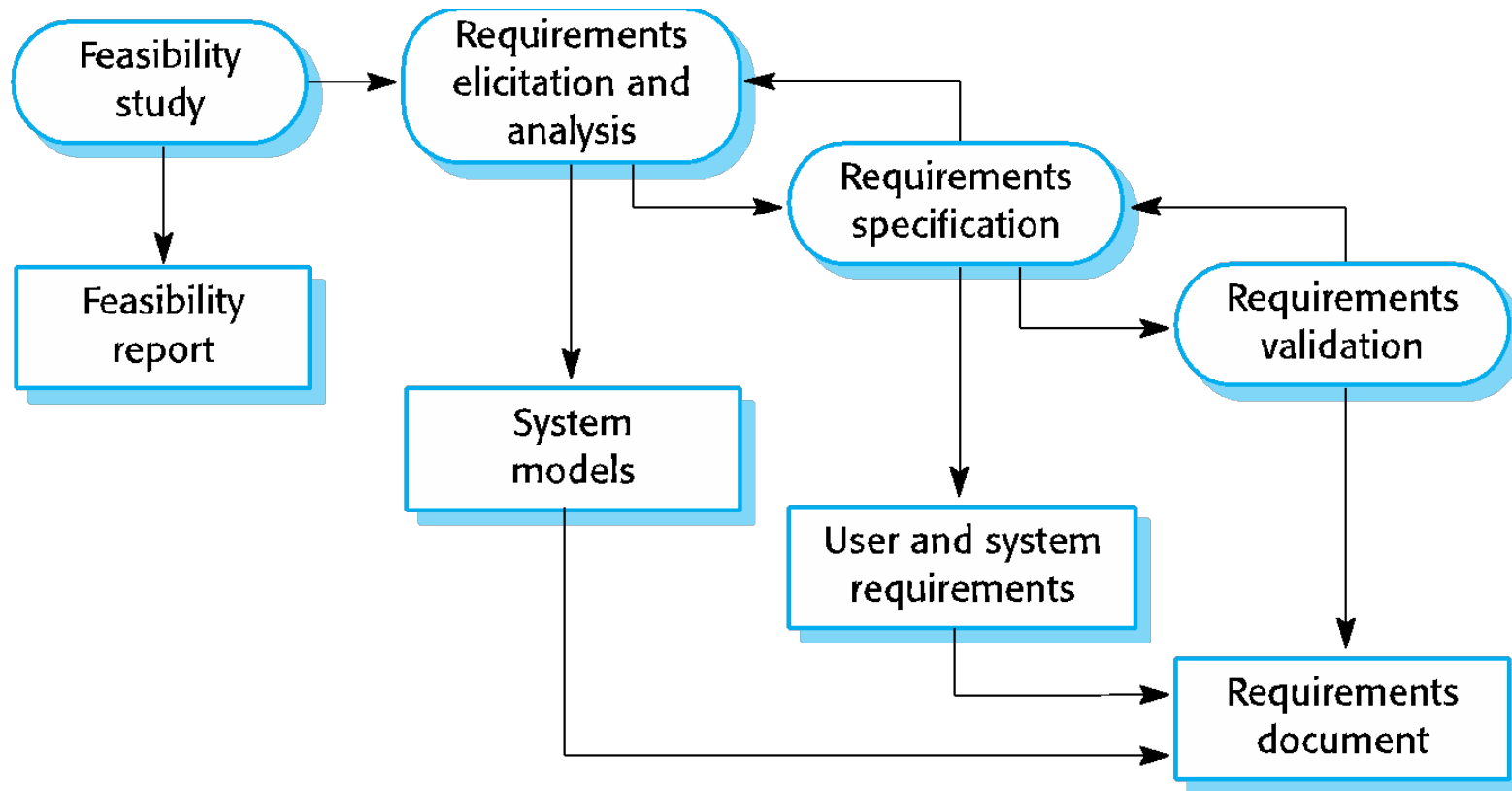
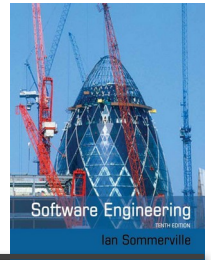
- ✧ Real software processes are inter-leaved sequences of technical, collaborative and managerial activities with the overall goal of specifying, designing, implementing and testing a software system.
- ✧ The four basic process activities of **specification**, **development**, **validation** and **evolution** are organized differently in different development processes.
- ✧ For example, in the waterfall model, they are organized in sequence, whereas in incremental development they are interleaved.

Software specification

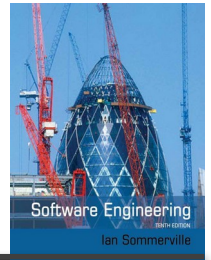


- ✧ The process of establishing what services are required and the constraints on the system's operation and development.
- ✧ Requirements engineering process
 - **Feasibility study**
 - Is it technically and financially feasible to build the system?
 - **Requirements elicitation and analysis**
 - What do the system stakeholders require or expect from the system?
 - **Requirements specification**
 - Defining the requirements in detail
 - **Requirements validation**
 - Checking the validity of the requirements

The requirements engineering process

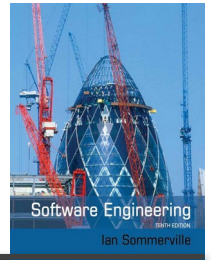


Software design and implementation



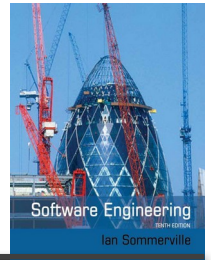
- ✧ The process of converting the system specification into an executable system.
- ✧ Software design
 - Design a software structure that realises the specification;
- ✧ Implementation
 - Translate this structure into an executable program;
- ✧ The activities of design and implementation are closely related and may be inter-leaved.

Software validation



- ✧ Verification and validation (V & V) is intended to show that a system conforms to its specification and meets the requirements of the system customer.
- ✧ Involves checking and review processes and system testing.
- ✧ System testing involves executing the system with test cases that are derived from the specification of the real data to be processed by the system.
- ✧ Testing is the most commonly used V & V activity.

Testing stages



✧ Development or component testing

- Individual components are tested independently;
- Components may be functions or objects or coherent groupings of these entities.

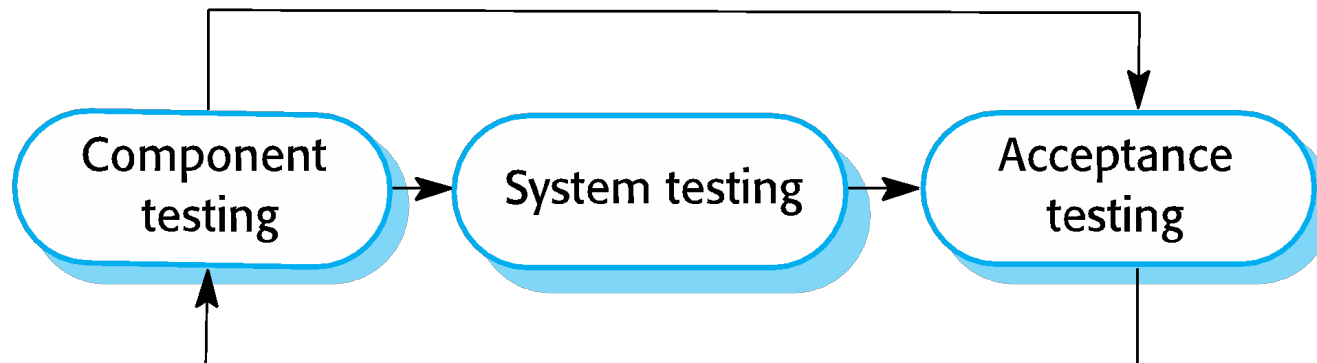
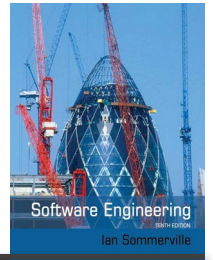
✧ System testing

- Testing of the system as a whole. Testing of emergent properties is particularly important.

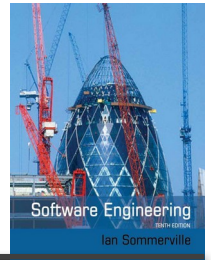
✧ Acceptance testing

- Testing with customer data to check that the system meets the customer's needs.

Stages of testing

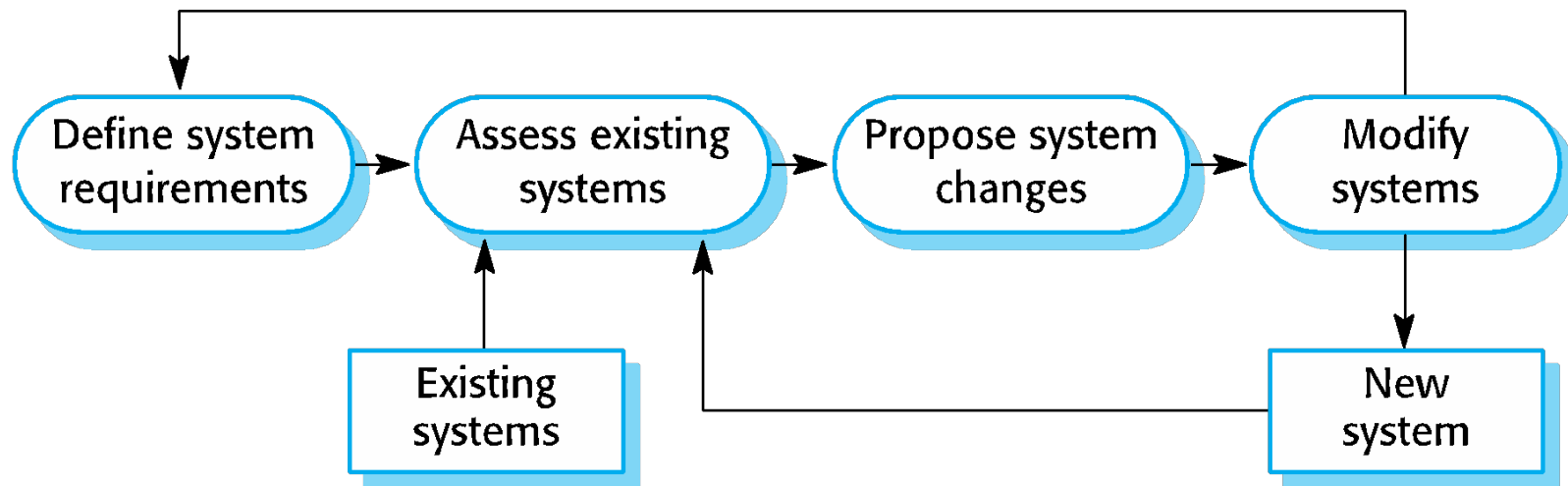
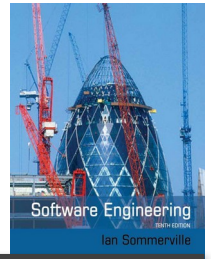


Software evolution

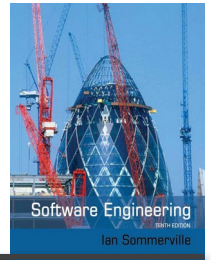


- ✧ Software is inherently flexible and can change.
- ✧ As requirements change through changing business circumstances, the software that supports the business must also evolve and change.
- ✧ Although there has been a demarcation between development and evolution (maintenance) this is increasingly irrelevant as fewer and fewer systems are completely new.

System evolution

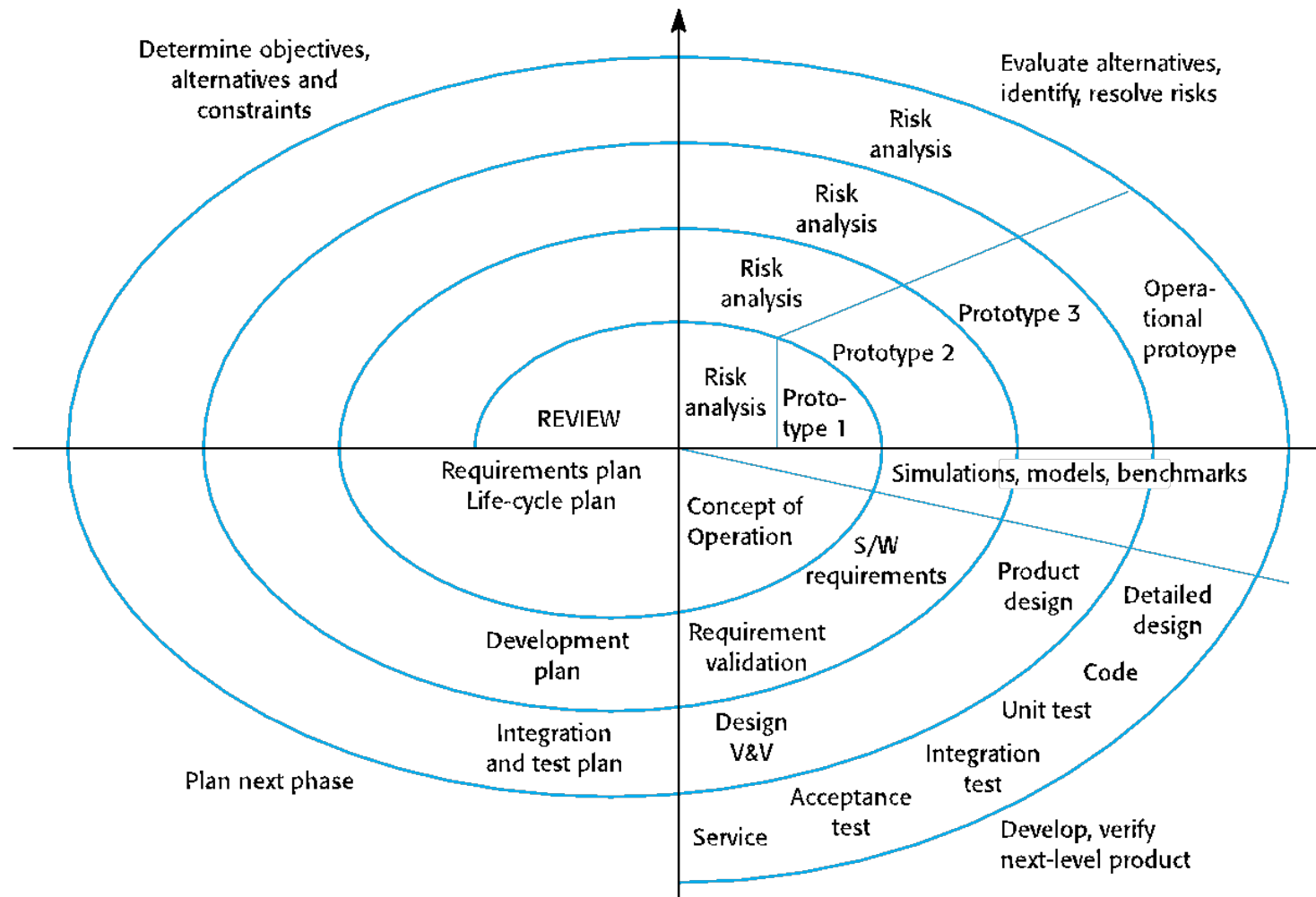


Boehm's spiral model

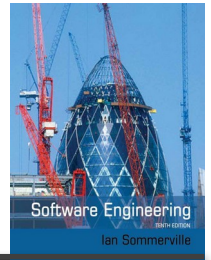


- ✧ Process is represented as a spiral rather than as a sequence of activities with backtracking.
- ✧ Each loop in the spiral represents a phase in the process.
- ✧ No fixed phases such as specification or design - loops in the spiral are chosen depending on what is required.
- ✧ Risks are explicitly assessed and resolved throughout the process.

Boehm's spiral model of the software process



Spiral model sectors



✧ Objective setting

- Specific objectives for the phase are identified.

✧ Risk assessment and reduction

- Risks are assessed and activities put in place to reduce the key risks.

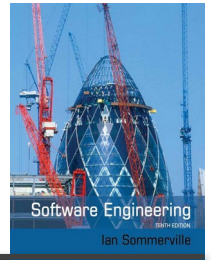
✧ Development and validation

- A development model for the system is chosen which can be any of the generic models.

✧ Planning

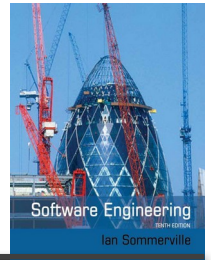
- The project is reviewed and the next phase of the spiral is planned.

Spiral model usage



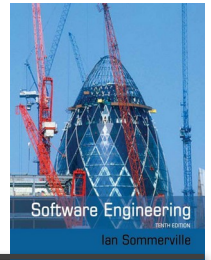
- ✧ Spiral model has been very influential in helping people think about iteration in software processes and introducing the risk-driven approach to development.
- ✧ In practice, however, the model is rarely used as published for practical software development.

Key points



- ✧ Software processes are the activities involved in producing a software system. Software process models are abstract representations of these processes.
- ✧ General process models describe the organization of software processes.
 - Examples of these general models include the 'waterfall' model, incremental development, and reuse-oriented development.
- ✧ Requirements engineering is the process of developing a software specification.

Key points



- ✧ Requirements engineering is the process of developing a software specification.
- ✧ Design and implementation processes are concerned with transforming a requirements specification into an executable software system.
- ✧ Software validation is the process of checking that the system conforms to its specification and that it meets the real needs of the users of the system.
- ✧ Software evolution takes place when you change existing software systems to meet new requirements. The software must evolve to remain useful.